

Pulse

Customer journey, experiment, target selection, and economic decision

A formal report from a reproducible synthetic digital-banking analytics run

Population	100,000 synthetic customers.
Observation period	2025-01-01 through 2026-01-25.
Primary decision	Continue controlled activation testing. Do not launch positive-contact outreach under the stated scenario assumptions.
Evidence classes	Synthetic descriptive metrics; randomized intent-to-treat estimate; observational DiD and IPTW diagnostics; temporal predictive evaluation; scenario economics.

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1 Executive summary and decision

Pulse is a synthetic, deterministic customer-analytics case study. It links a customer journey, cohort retention, an activation experiment, an observational campaign analysis, churn and response models, monitoring outputs, and targeting economics. The analysis does not use customer data.

The primary decision is narrow. Continue the activation intervention as a controlled experiment because the randomized estimate is positive and precise for the stated eligible population. Do not launch a positive-contact campaign under the supplied fictional value and cost assumptions: every positive-contact strategy has negative expected net value, while the expected-value policy contacts no customers.

Table 1: Decision-relevant results from the frozen full run.

Evidence	Recorded result and interpretation
Journey	Overall D30 retention is 23.604% (23,604 customers). Sustained sequential D30 engagement is 20.909% (20,909 customers). These are different metrics.
Randomized experiment	Treatment minus control first-week activation risk difference is +9.16 pp; the 95% confidence interval is +8.28 pp to +10.05 pp; $p < 0.001$.
Observational campaign	Unweighted DiD is +1.03 pp. Stabilized IPTW DiD is +0.95 pp with 95% interval +0.09 pp to +1.80 pp. Diagnostics do not validate a causal effect.
Prediction	The selected regularized logistic churn model records ROC AUC 0.752 and top-decile lift 2.673 on a later temporal holdout. This is predictive association, not treatment benefit.
Economics	The best recorded policy is “expected value.” Its expected net value is 0 fictional units after evaluating 11,728 customers.

The report separates verified facts from interpretation. All metrics in this section come from `artifacts/results.json` and its linked tables. The decision is an interpretation conditional on those metrics, the synthetic data-generating process, and the documented assumptions.

2 Synthetic-data scope and metric contract

The full run contains 100,000 synthetic customers from 2025-01-01 through 2026-01-25. The simulation includes onboarding, funding, meaningful transactions, product activity, cohort changes, a campaign, experiment eligibility, and subsequent outcomes. Public Indian-bank disclosures inform lifecycle structure only; they do not calibrate the numerical retention rates.

The canonical activation definition is: *Funded and at least one settled meaningful transaction in signup calendar days 0-7 inclusive..* A meaningful transaction is a settled, non-negative transaction whose type is `card_transaction`, `transfer`, or `bill_payment`. Retention uses day windows: D7 is days 7–13, D30 is days 30–36, D60 is days 60–66, and D90 is days 90–96. A funnel row uses the immediately preceding stage as its denominator unless it is labelled as a signup conversion.

Table 2: Sequential journey counts and denominators.

Stage	Customers	Signup conversion	Previous-stage conversion
signup	100,000	100.0%	100.0%
onboarding completed	74,458	74.5%	74.5%
account funded	58,078	58.1%	78.0%
first transaction 7d	42,254	42.3%	72.8%
repeat usage	40,996	41.0%	97.0%
sustained engagement d30	20,909	20.9%	51.0%

Evidence: `artifacts/results.json`, `artifacts/tables/journey.csv`, `config/metrics.yml`, and `docs/metric_definitions.md`.

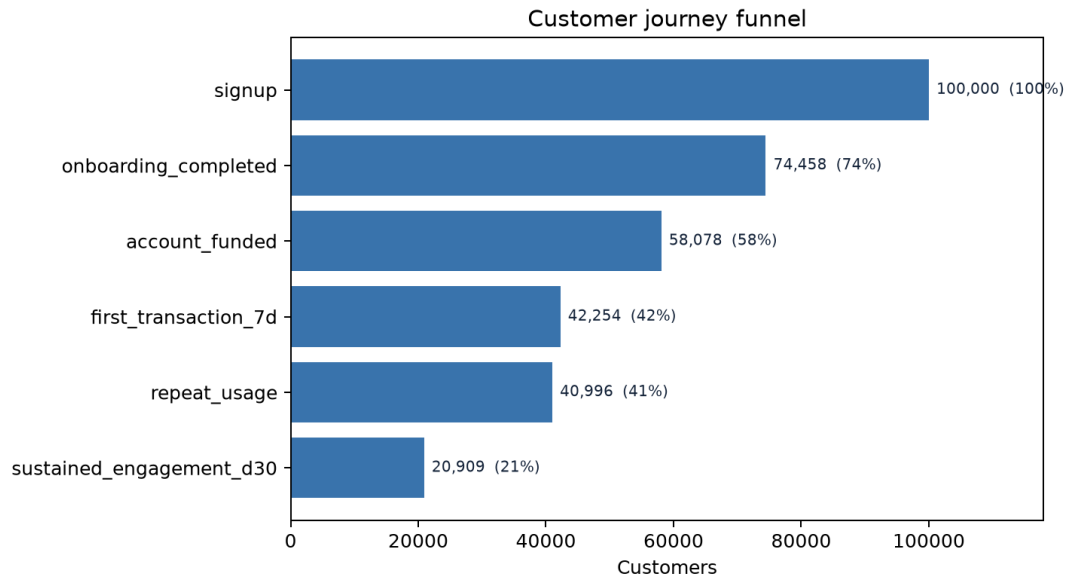


Figure 1: Journey funnel from the generated table. The sustained sequential D30 row is not the same denominator as overall D30 retention.

3 Journey and cohorts

The early journey narrows from signup to onboarding, funding, and first meaningful transaction. The first transaction in the signup-week window is the activation outcome used in the randomized experiment. Repeat usage is a sequential stage. Sustained D30 engagement is a stricter sequential condition than overall D30 retention, which is measured directly against signup.

The mature-cohort estimates decline from 29.7% at D7 to 23.6% at D30, 20.5% at D60, and 16.3% at D90. The cohort table stores maturity windows and the analysis cutoff, so recent cohorts are not silently treated as inactive.

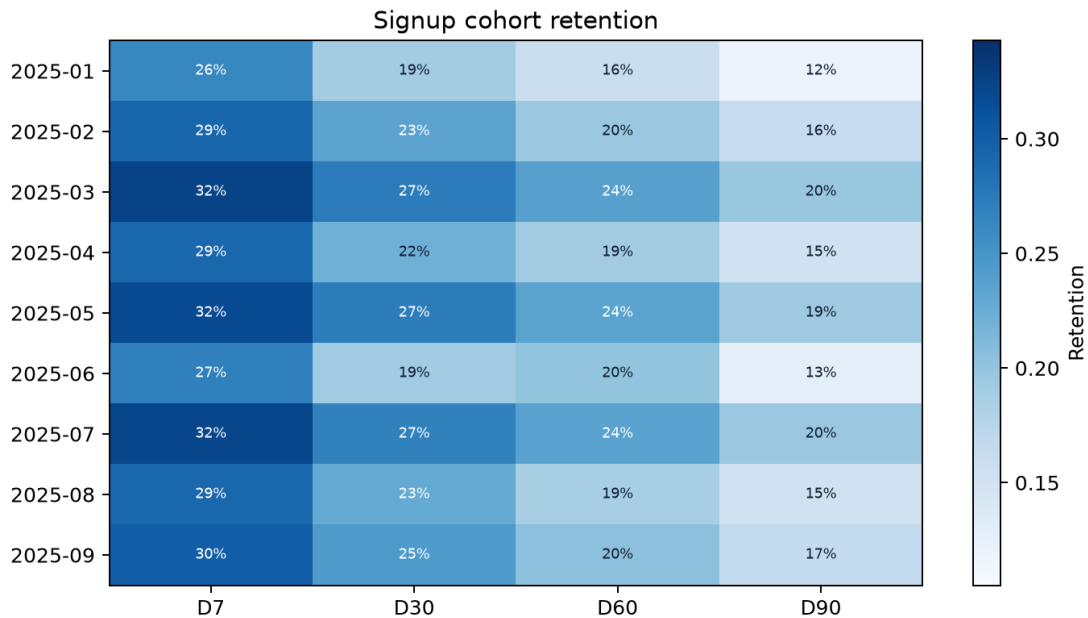


Figure 2: Retention by signup cohort and observation horizon. The plotted values are descriptive synthetic cohort estimates.

The June 2025 cohort shows a reactivation bump from its D30 window to its D60 window: the table records 19.18% at D30 and 20.02% at D60. This increase is compatible with later activity after an earlier inactive window. It does not establish why the bump occurred.

Claim boundary. Cohort retention identifies where activity differs by window and signup month. It does not identify an intervention effect or explain the June bump.

4 Randomized experiment

4.1 Design and estimand

The experiment asks whether an onboarding nudge or incentive increases first-week activation among customers funded by signup day 3 and not activated at eligibility. Assignment precedes the outcome. The primary estimand is the intent-to-treat risk difference, treatment minus control, for eligible customers.

The frozen table contains 18,647 control customers and 18,693 treatment customers. The control activation rate is 69.64% and the treatment activation rate is 78.80%. The normal-approximation minimum detectable effect is +1.33 pp. Balance and guardrail outputs are reported as integrity checks, not as replacements for the outcome estimate.

Table 3: Experiment arms and secondary retention outcomes.

Arm	Customers	Activations	D7 activation	D30 retention	Sustained D30
Control	18,647	12,985	69.6%	39.6%	34.5%
Treatment	18,693	14,730	78.8%	42.7%	39.0%

4.2 Effect and guardrails

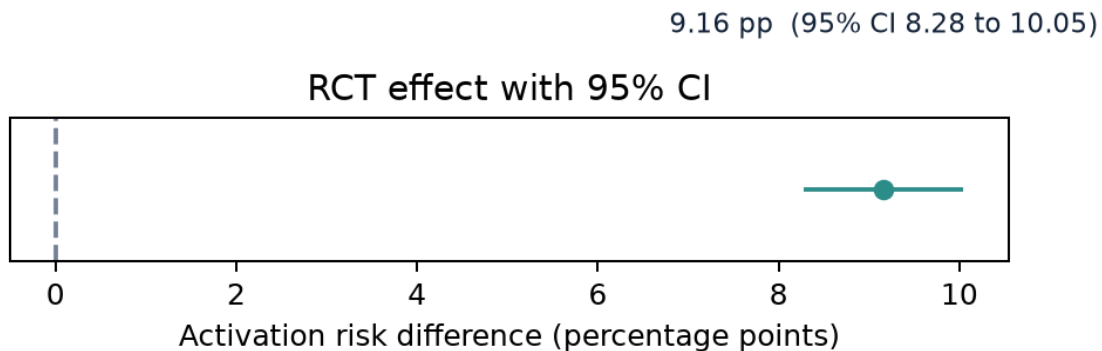


Figure 3: Randomized activation effect and confidence interval from the frozen experiment summary.

The ITT risk difference is +9.16 pp with a 95% confidence interval of +8.28 pp to +10.05 pp and $p < 0.001$. The p-value is reported as a threshold because the stored floating-point value is zero, not because a literal zero probability was established. The interval is far from zero under the stated experiment model, but statistical evidence does not by itself establish profit, durable retention, or external validity.

Table 4: Experiment guardrails recorded in the frozen run.

Guardrail	Value	Threshold	Status
sample ratio mismatch	0.501	0.020	pass
max absolute smd	0.023	0.100	pass
support contact rate difference	0.019	0.050	pass
failed payment rate difference	-0.005	0.030	pass
incentive cost per customer	0.350	0.500	pass

Evidence: `artifacts/results.json`, `artifacts/tables/experiment_summary.csv`, `artifacts/tables/rct_guardrails.csv`, and `docs/experiment_design.md`.

5 Observational campaign: DiD and IPTW

The campaign analysis uses a selected-region observational design. Its repeated weekly panel contains 21,275 comparison customers and 28,540 selected-region customers. It reports a difference-in-differences contrast and a stabilized inverse-probability-of-treatment weighted (IPTW) estimate. The unweighted DiD is +1.03 pp; the weighted estimate is +0.95 pp with standard error 0.0044 and 95% interval +0.09 pp to +1.80 pp. The effective sample size is 42,116 under the recorded uncertainty calculation.

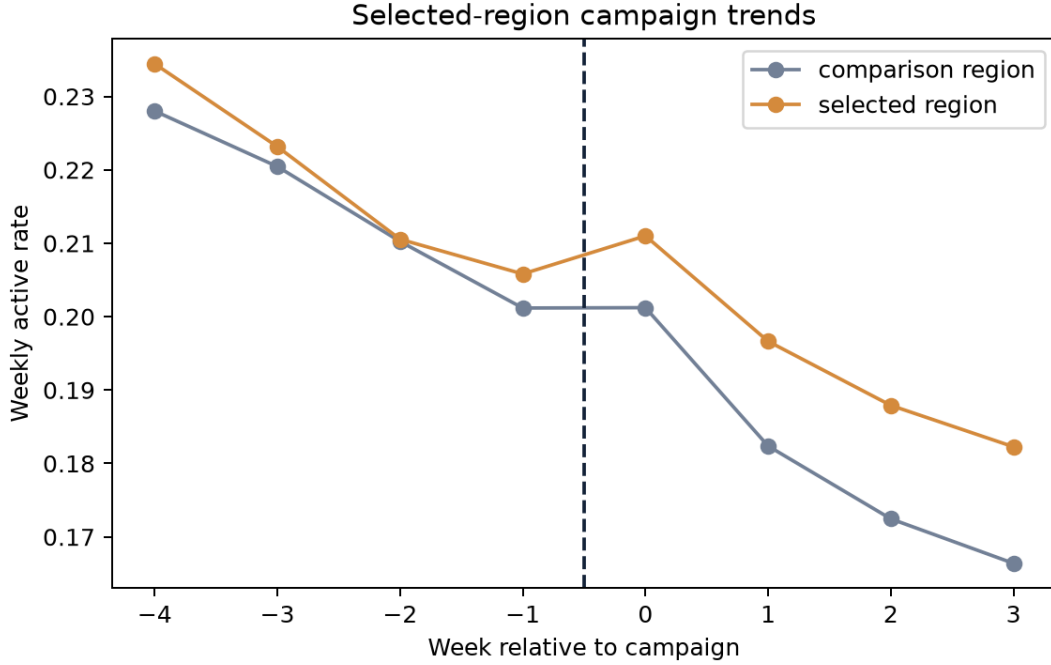


Figure 4: Weekly activity rates for the selected region and comparison group. The post-period gap is larger than the recorded pre-period gaps.

Table 5: Pre-period and post-period activity rates. Relative week 0 is the first post-period row in the generated table.

Relative week	Comparison	Selected region	Difference
-4	22.8%	23.4%	+0.64 pp
-3	22.0%	22.3%	+0.27 pp
-2	21.0%	21.1%	+0.03 pp
-1	20.1%	20.6%	+0.46 pp
0	20.1%	21.1%	+0.98 pp
1	18.2%	19.7%	+1.43 pp
2	17.2%	18.8%	+1.55 pp
3	16.6%	18.2%	+1.59 pp

5.1 Diagnostics and assumptions

The common-support diagnostic records 100.0% and the maximum absolute standardized mean difference after stabilized weighting is 0.00349. The unweighted region indicators remain highly imbalanced; weighting reduces measured differences in the recorded covariates. These are diagnostics only. The registry status “pass” means the configured diagnostic thresholds passed. It is not causal validation.

The estimate requires parallel untreated trends, no differential contemporaneous shock, stable composition, exchangeability conditional on measured covariates, positivity, correct propensity specification, and no post-treatment features. The stored standard error uses a normal approximation on weighted customer-week means and ignores within-customer correlation. Region assignment is a design feature that must be justified with internal evidence before treating the estimate as causal.

Claim boundary. The campaign result is an observational estimate subject to its assumptions. Measured

balance and overlap do not address unmeasured confounding or prove that the campaign caused the post-period difference.

6 Churn and response models

The churn task predicts no meaningful activity in the following 30 days for customers active at the feature cutoff. Regularized logistic is fixed as the selected operational model for interpretability; the comparison table remains the evaluation record. It is evaluated on a later temporal holdout of 11,728 customers. Its ROC AUC is 0.752, top-decile lift is 2.673, and top-decile capture is 26.7%. The top-decile set contains 1,173 customers. These values support a capacity-aware risk ranking under the evaluation sample. They do not estimate treatment benefit.

Table 6: Churn model comparison on the later temporal holdout.

Model	Brier	ROC AUC	PR AUC	Lift at 10%	Capture at 10%
rule baseline	0.158	0.500	0.196	0.995	10.0%
regularized logistic	0.133	0.752	0.439	2.673	26.7%
hist gradient boosting	0.133	0.750	0.435	2.669	26.7%

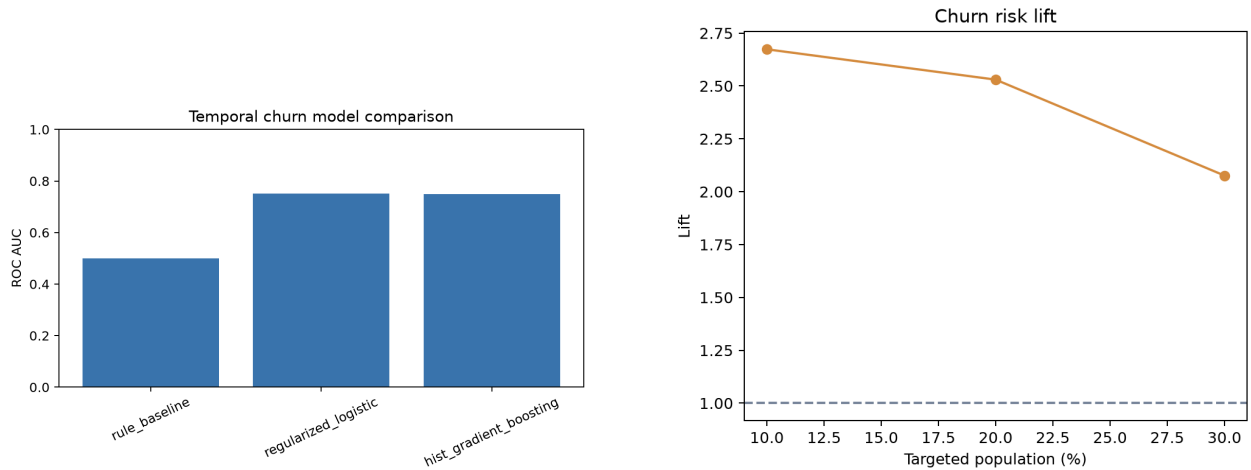


Figure 5: Model comparison and concentration of churn events in the scored population.

The response propensity model predicts observed activation among treated, eligible customers. It is evaluated on a later temporal holdout of 4,635 customers. Its ROC AUC is 0.607, top-decile lift is 1.160, and top-decile capture is 11.6%. The top-decile set contains 464 customers. This target is not an uplift estimate: it does not compare each customer's treated and untreated potential outcomes. A contact policy therefore requires incremental response evidence, a holdout design, and harm and cost guardrails.

7 Targeting economics

The economic policy evaluates 11,728 customers under explicit fictional assumptions: conditional treatment effect 12.0%, contact cost 0.25 units, incentive cost 0.50 units, and a capacity fraction of 10%. The strategy table records expected net value under these assumptions.

Table 7: Targeting strategy comparison. Expected values are fictional units.

Strategy	Customers contacted	Contact rate	Expected net value	Profitable
contact all	11,728	100.0%	-8,567	No
risk	1,172	10.0%	-815	No
value	1,172	10.0%	-859	No
response	1,172	10.0%	-855	No
expected value	0	0.0%	0	No

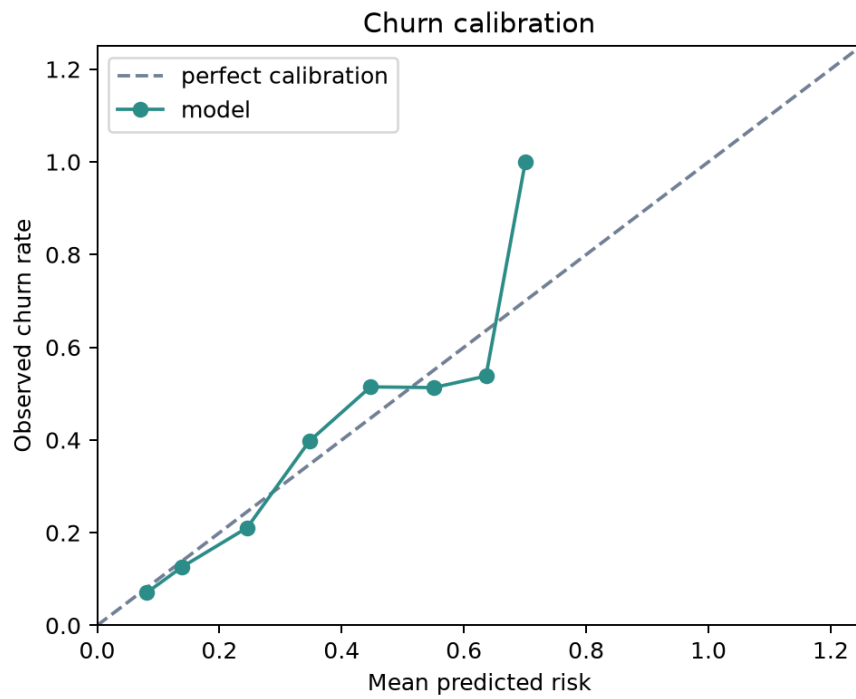


Figure 6: Calibration by predicted-risk bin for the selected churn model. Sparse high-risk bins remain a limitation.

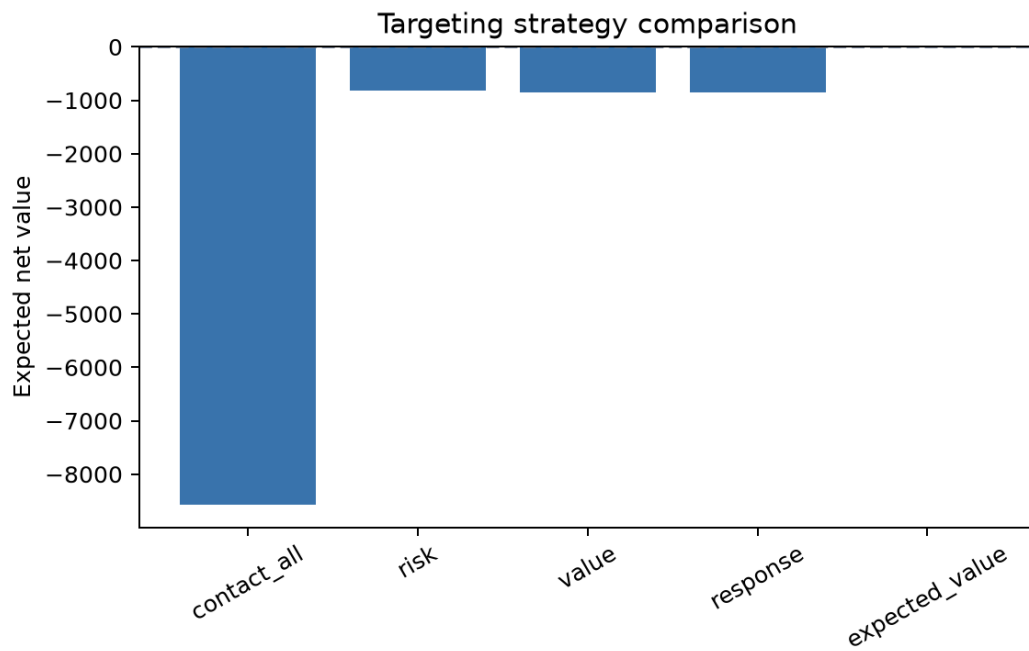


Figure 7: Expected net value by targeting strategy under the recorded scenario assumptions.

The expected-value policy is therefore conservative: it contacts no customers because the estimated value of a positive contact is not sufficient under the scenario. This is a scenario decision, not evidence that all real outreach is unprofitable. The assumptions require internal value, cost, incremental-effect, and capacity inputs before operational use.

8 Monitoring and temporal stability

The monitoring artifact reports performance by temporal bucket and feature population stability index (PSI). The maximum feature PSI is 0.0023, below the configured low-drift reference of 0.1. This is a current synthetic-run diagnostic. It does not establish production stability.

Table 8: Temporal holdout performance for the selected churn model.

Time bucket	Customers	Base rate	ROC AUC	Lift at 10%	Calibration error
2025-08-01	1,668	15.8%	0.761	3.405	0.024
2025-09-01	5,318	21.5%	0.738	2.477	0.002
2025-10-01	4,742	18.8%	0.761	2.793	0.015

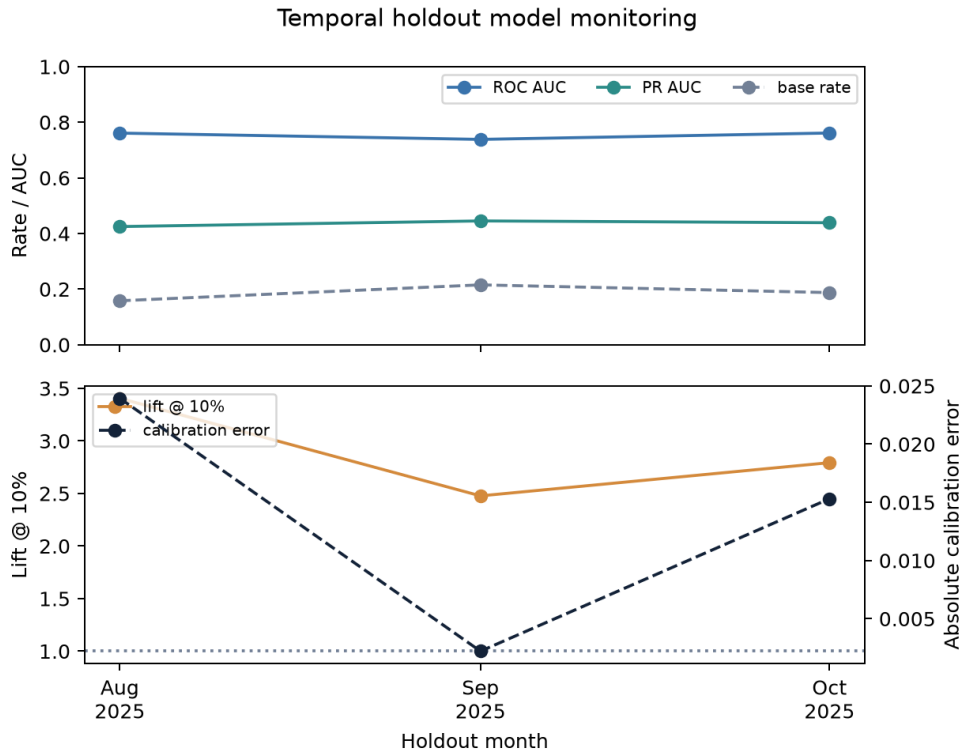


Figure 8: Performance over temporal holdout buckets. Monitoring should retain the metric definitions and operating-capacity thresholds used here.

The observed bucket variation is material enough to justify continued monitoring, even though the stored status does not indicate recalibration or retraining. A production monitoring plan would need a live data-quality contract, missingness and latency checks, outcome-delay handling, subgroup harm checks, and an approved response to threshold breaches.

9 Indian-bank directional evidence

The simulator uses public disclosures from Indian banks to motivate lifecycle heterogeneity. The disclosures describe different units, such as registered users, monthly active users, engagements, digital transaction share, and product adoption. They are not comparable to the Pulse signup-cohort retention windows.

Table 9: Directional public evidence and design implications. These sources do not provide numerical benchmarks for Pulse retention.

Bank	Period	Evidence recorded in the source registry	Design implication
State Bank of India	FY2025	YONO had 8.8 crore registered users, up from 7.4 crore in March 2024; retail internet banking had 13.9 crore customers.	Model multiple digital channels and changing adoption mixes.
Axis Bank	FY2025	Mobile banking had about 3 crore registered customers and more than 15 million monthly active users; the open platform covered 30+ products.	Separate registration, monthly activity, and product depth.
HDFC Bank	FY2025	The bank reported 3.7 crore+ monthly customer engagements on HDFC Bank One, 97% of financial transactions through digital channels, and 19.3 lakh merchants onboarded to SmartHub Vyapar.	Use heterogeneous retail and merchant activity intensity.
ICICI Bank	February 2024	iMobile had onboarded more than one crore customers of other banks, with repeat use across payments, transfers, recharges, and balance checks.	Include non-primary users and product-specific repeat behaviour.
Kotak Mahindra Bank	FY2025	A regulatory direction paused digital onboarding and new credit-card issuance until February 2025 while the bank deepened existing 811 relationships and reworked its persona-based app strategy.	Include cohort-level acquisition shocks and reactivation.

The evidence supports modeling distinct registered and active states, changing channel mix, product depth, non-primary users, repeat activity, and cohort-level shocks. It does not support setting the synthetic D7, D30, D60, or D90 rates equal to any reported bank metric.

10 Conclusions and limitations

The verified results support three limited conclusions. First, the activation experiment records a positive ITT estimate for the eligible synthetic population. Second, the selected-region campaign produces a positive observational contrast, and the configured overlap and balance diagnostics pass, but the causal assumptions remain unverified. Third, the churn ranking is useful as predictive concentration evidence, while response scores and targeting economics do not establish incremental value.

The resulting decision is to keep activation behind controlled experimentation and to withhold positive-contact outreach under the supplied fictional assumptions. The report does not support a production rollout, a claim of customer impact, or a claim that the campaign caused the observed change.

Limitations are structural. The data are behaviorally plausible simulation, not customer data. The simulation’s hidden mechanisms can validate selected calculations but do not validate external performance. The observational analysis can remain confounded. Temporal evaluation covers the recorded holdout buckets but cannot forecast policy or population change. Predictive models can drift and can harm subgroups even when aggregate metrics look stable. The economics are assumption-driven. External bank disclosures are directional and use incompatible denominators.

A Reproducibility and evidence map

The report is generated in two steps. The evidence generator reads the frozen artifacts and writes a LaTeX include. Tectonic compiles the main source to the requested PDF path.

```
make report
```

The explicit commands are:

```
.venv/bin/python scripts/build_latex_inputs.py
tectonic -X compile --outdir reports \
  reports/pulse_case_study.tex
```

The generator validates required denominators before writing the include. It reconciles the signup and D30 counts, each experiment arm’s counts and rate, the randomized effect and interval, campaign group sizes, selected-model and response metrics, holdout populations, and the weighted maximum standardized mean difference across the manifest and detailed tables.

Table 10: Primary report inputs and their roles.

Artifact	Role in this report
<code>artifacts/results.json</code>	Run scope, headline metrics, claim scopes, assumptions, and limitations.
<code>artifacts/tables/*.csv</code>	Journey, cohorts, experiment, guardrails, campaign diagnostics, models, monitoring, and economics rows.
<code>artifacts/figures/*.png</code>	Generated figures embedded without recomputation.
<code>config/external_benchmarks.yml</code>	Directional Indian-bank evidence and source URLs.
<code>config/metrics.yml</code> and <code>docs/metric_definitions.md</code>	Metric definitions and window contracts.

B References

- [1] State Bank of India, *SBI FY25 Analyst Presentation*, slide 46. [source link](#)
- [2] Axis Bank, *Integrated Annual Report 2024–25*, p. 127. [source link](#)
- [3] HDFC Bank, *Integrated Annual Report 2024–25*, pp. 7 and 140–141. [source link](#)
- [4] ICICI Bank, “iMobile is used by one crore customers from other banks,” February 2024. [source link](#)
- [5] Kotak Mahindra Bank, *Integrated Annual Report 2024–25*, pp. 9 and 24–25. [source link](#)